EDGE-AI ACOUSTIC MONITORING

Acoustic Edge-AI Pipe Leak & Micro-Crack Detector

Real-time, on-device detection of pipe leaks and micro-cracks using an ESP32, an I2S MEMS sensor, and a TinyML acoustic classifier.



Capstone / IoT Prototype Project

THE CHALLENGE

Problem Statement

Weak, hard-to-catch signals

Small leaks and micro-cracks produce faint acoustic signatures that are easy to miss during manual inspection.

Noisy industrial environments

Background noise from pumps, machinery, and surroundings masks the acoustic cues inspectors rely on.

Specialized equipment needed

Conventional inspection methods often require costly tools and trained acoustic experts on site.

Need for intelligent detection

A compact, real-time, intelligent system is needed to flag abnormal pipe sounds automatically.

OUR APPROACH

Proposed Solution

1 Capture

An I2S MEMS sensor captures acoustic signals directly from the pipe surface.

2 Process

The ESP32 receives the audio stream through its digital I2S interface.

3 Classify

A TinyML model analyzes the acoustic pattern locally, right on the ESP32.

4 Alert

On detecting a leak-related signal, the OLED and buzzer raise an instant alert.

Why it works

- On-device inference removes dependence on constant cloud connectivity.
- I2S digital audio avoids analog noise pickup, improving signal fidelity.
- A pipe clamp mount improves acoustic coupling and measurement consistency.
- OLED + buzzer give the user a result they can see and hear instantly.

OUTCOME

A compact, portable prototype that flags abnormal pipe acoustics in real time — no specialist or constant connectivity required.

HOW IT FITS TOGETHER

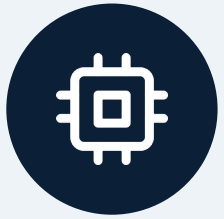
System Architecture



The pipe produces acoustic vibrations from normal flow, leakage or defects. The I2S sensor captures these signals and streams them to the ESP32, which classifies the acoustic condition with an on-device TinyML model — then drives the OLED and buzzer to alert the user.

HARDWARE

Hardware Components (1/2)



ESP32 Development Board

- Main controller of the system
- Receives digital audio via I2S
- Runs the TinyML model locally
- Drives OLED & buzzer outputs



I2S MEMS Sensor

- Captures acoustic pipe vibrations
- Digital I2S interface to ESP32
- Picks up hissing / leak patterns
- Streams audio for processing



0.96" OLED Display

- Shows system operating status
- Displays condition & AI confidence
- Real-time visual feedback
- Main visual output of the prototype

HARDWARE

Hardware Components (2/2)



Active Buzzer Module

- Audible warning on leak detection
- Triggers past confidence threshold
- Alerts without watching the screen
- Main alarm / notification component



3D-Printed / Acrylic Pipe Clamp

- Holds sensor firmly against the pipe
- Improves acoustic coupling
- Reduces movement, adds consistency
- Customizable to different pipe sizes



Jumper Wires & Breadboard

- Connects components while prototyping
- Links ESP32, sensor, OLED & buzzer
- Enables fast testing & iteration
- Replaceable later with a custom PCB

TOOLCHAIN

Software & AI Technologies



Arduino IDE / PlatformIO

Writes and uploads C++ firmware to the ESP32; handles sensor, OLED and buzzer operations and integrates the trained TinyML model.



Edge Impulse

Collects and prepares the acoustic dataset, extracts features such as MFCCs, trains the model, and deploys it to the ESP32.

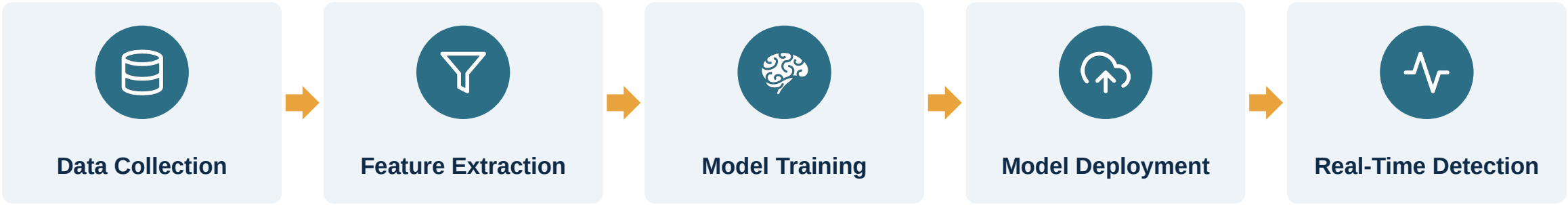


Edge Impulse Data Forwarder / CLI

Transfers acoustic samples from the ESP32 to Edge Impulse and supports building labeled real-world training datasets.

WORKFLOW & GOALS

AI / Detection Process & Objectives



Data is collected for normal vs. leak/defect sounds, MFCC features are extracted, a model is trained and deployed to the ESP32 for real-time, on-device detection.

PROJECT OBJECTIVES

1. Detect acoustic patterns linked to pipe leaks and defects.
2. Use Edge AI / TinyML for real-time on-device detection.
3. Reduce dependence on cloud connectivity during detection.
4. Provide immediate, easy-to-understand alerts.
5. Develop a compact and portable prototype.